

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

Code of Conduct:

192511445

I SHAIK KHATAM RIZHAN (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:







Annexure A



Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.

3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

		context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

CO2-AT 2

SCENARIO BASED QUESTIONS

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COURSE CODE: CSA0702

COURSE NAME: COMPUTER NETWORKS.

1. Given:

IP Address Block: $192.168.100.0/24$

Subnet Allocation:

⇒ School of computing $\rightarrow /25$

⇒ School of electronics $\rightarrow /26$

⇒ School of Mechanical engineering $\rightarrow /27$

Calculation:

School of computing ($/25$)

⇒ Subnet Mask: $255.255.255.128$

⇒ Total Addresses: $2^1 (32 - 25) = 128$

⇒ Usable hosts: $128 - 2 = 126$

Network Address: $192.168.100.0$

Host Range: $192.168.100.1 - 192.168.100.126$

Broadcast Address: $192.168.100.127$

School of electronics ($/26$)

⇒ Subnet Mask: $255.255.255.192$

⇒ Total Addresses: 64

⇒ Usable hosts: 62

Network Address : 192.168.100.128

Host Range : 192.168.100.129 - 192.168.100.190

Broadcast Address : 192.168.100.191

School of Mechanical Engineering (127)

⇒ Subnet Mask : 255.255.255.224

⇒ Total Addresses : 32

⇒ usable Hosts : 30

Network Address : 192.168.100.192

Host Range : 192.168.100.193 - 192.168.100.222

Broadcast Address : 192.168.100.223

Remaining unused IP Address Range.

192.168.100.224 - 192.168.100.255

Result :

Department	Network Address	Host Range	Broadcast Address	Usable Hosts
School of computing	192.168.100.0/25	192.168.100.1 - 192.168.100.126	192.168.100.127	126
School of electronics	192.168.100.128/26	192.168.100.129 - 192.168.100.190	192.168.100.191	62
School of Mechanical engineering	192.168.100.192/27	192.168.100.193 - 192.168.100.222	192.168.100.223	30

2, Given :

IP Address Block : 10.10.0.0/16

Subnet Allocation

⇒ Software Development : → /24

⇒ Quality Assurance → /25

⇒ Human Resource → /26

⇒ Executive Management → /27

Remaining unused IP Address Range.

10.10.1.224 - 10.10.255.255

Result Table :

Department	Network Address	Host Range	Broadcast Address	usable Hosts
Software development	10.10.0.0/24	10.10.0.1 - 10.10.0.254	10.10.0.255	254
Quality Assurance	10.10.1.0/25	10.10.1.1 - 10.10.1.126	10.10.1.127	126
Human Resources	10.10.1.128/26	10.10.1.129 - 10.10.1.190	10.10.1.191	62
Executive Management	10.10.1.192/27	10.10.1.193 - 10.10.1.222	10.10.1.223	30

3. Given

IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000 /64

Solution

Compressed Form

2001:db8:abcd::/64

Expanded Form

2001:0db8:abcd:0000:0000:0000:0000:0000 /64

Network prefix:

Since the prefix length is /64, the first 64 bits represent the network portion.

Network prefix: 2001:db8:abcd:0000::/64

Calculation:

Software Development (124)

Subnet Mask: 255.255.255.0

Total Addresses = 256

usable hosts = 254

Network Address: 10.10.0.0

Hosts Range: 10.10.0.1 - 10.10.0.254

Broadcast Address: 10.10.0.255

Quality Assurance (125)

Network Address: 10.10.1.0

Hosts Range: 10.10.1.1 - 10.10.1.126

Broadcast Address : 10.10.1.127
usable Hosts : 126

Human Resources (126)

Network Address : 10.10.1.128

Host Range : 10.10.1.129 - 10.10.1.190

Broadcast Address : 10.10.1.191

usable Hosts : 62

Executive Management (127)

Network Address : 10.10.1.192

Host Range : 10.10.1.193 - 10.10.1.222

Broadcast Address : 10.10.1.223

usable Hosts : 30

Total Host Addresses :

Host Bits = $128 - 64 = 64$

Total Hosts = $2^{64} = 18,446,744,073,709,551,616$

Conclusion :

The compressed IPv6 address is 2001 db8 : abcd ::
164 and the subset can support 2^{64} host addresses,
making IPv6 suitable for very large networks.

4, Given Routing Table.

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP Address : 192.168.2.165

Solution:

The destination IP 192.168.2.45 belongs to the subnet 192.168.2.0/24

Therefore, the router selects the routing table entry.

192.168.2.0/24

The packet is forwarded through the interface connected to the 192.168.2.0/24 network.

If no routing table entry matches the destination IP, the router forwards the packet using the default Route (0.0.0.0/0).

Result:

⇒ Destination Subnet: 192.168.2.0/24

⇒ Matching Route: 192.168.2.0/24

⇒ Next-Hop: Interface connected to 192.168.2.0/24

⇒ Default Route: 0.0.0.0/0

5, Given:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

Administration LAN.

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Host Range: 192.168.10.1 - 192.168.10.62

Default Gateway: 192.168.10.1

Suggest computer IP Addresses.

PC1 → 192.168.10.2

PC2 → 192.168.10.3

PC3 → 192.168.10.4

Finance LAN:

Network Address : 192.168.10.64

Broadcast Address : 192.168.10.127

Host Range : 192.168.10.65 - 192.168.10.126

Default Gateway : 192.168.10.65

Suggest computer IP Addresses:

PC1 → 192.168.10.66

PC2 → 192.168.10.3

PC3 → 192.168.10.4

Finance LAN:

Network Address : 192.168.10.64

Broadcast Address : 192.168.10.127

Host Range : 192.168.10.65 - 192.168.10.126

Default Gateway : 192.168.10.65

Suggested computer IP Addresses:

PC1 → 192.168.10.66

PC2 → 192.168.10.67

PC3 → 192.168.10.68

Result Table :

LAN	Network Address	Host Range	Broadcast Address	Default Gateway.
Administration	192.168.10.0/26	192.168.10.1 - 192.168.10.62	192.168.10.63	192.168.10.1
Finance	192.168.10.64/26	192.168.10.65 - 192.168.10.126	192.168.10.127	192.168.10.65

conclusion :

The router uses 192.168.10.1 as the gateway for the Administration LAN and 192.168.10.65 as the gateway for the Finance LAN, enabling communication within and between the two LANs.

Q22